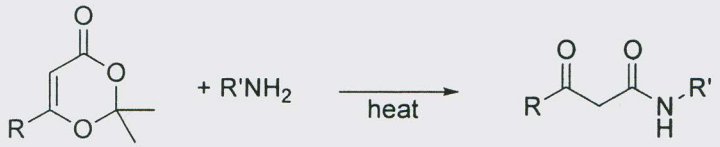
Organic Cume
December 3, 2010

In Monday's seminar, Professor Andy Phillips mentioned a number of interesting reactions. Answer the following questions about the examples below, taken from his seminar.

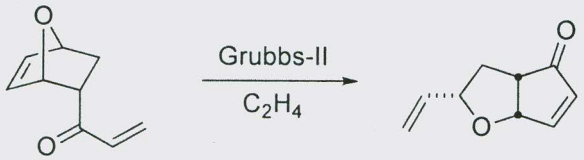
1. He described the reaction below as a retro-Diels-Alder fragmentation followed by nucleophilic addition to a reactive intermediate. Draw a mechanism for the transformation via the retro-Diels-Alder reaction, including the structure of the intermediate, and include any byproducts formed.



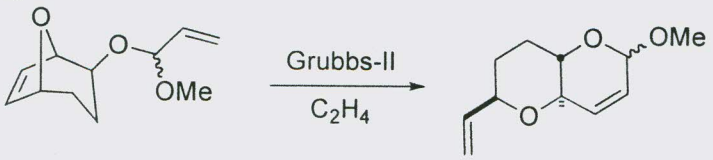
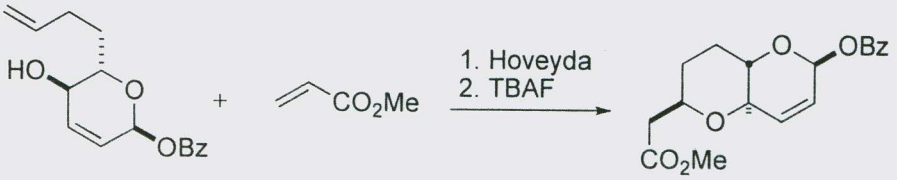
2. There is another mechanism that could also account for the formation of the observed products. Draw it.

3. He also showed several examples of olefin metathesis. Draw the intermediates of the following transformations. "Grubbs-II" and "Hoveyda" are metathesis catalysts whose structure is not needed to identify the intermediates in the sequences. "TBAF" is tetrabutyl ammonium fluoride.

a)



b)



c)